

ACR13024

## 2-Phenoxyethanol

### SECTION 1. IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

**产品说明:**  
**Product Description:** 乙二醇苯醚  
 2-Phenoxyethanol

**Cat No. :** 130240000; 130240010; 130240025; 130240051; 130240250; 130242500  
**Synonyms** Ethylene glycol monophenyl ether  
**CAS No** 122-99-6  
**Molecular Formula** C8 H10 O2

**Supplier**
**UK entity/business name**  
 Fisher Scientific UK  
 Bishop Meadow Road,  
 Loughborough, Leicestershire LE11 5RG, United Kingdom

**EU entity/business name**  
 Thermo Fisher Scientific  
 Janssen Pharmaceuticaaan 3a, 2440 Geel, Belgium

**Emergency Telephone Number** For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11  
 Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99  
**CHEMTREC** Tel. No. **US**:001-800-424-9300 / **Europe**:001-703-527-3887

**E-mail address** begel.sdsdesk@thermofisher.com

**Recommended Use** Laboratory chemicals.  
**Uses advised against** No Information available

### SECTION 2. HAZARD IDENTIFICATION

**Physical State**  
Liquid

**Appearance**  
Colorless

**Odor**  
aromatic

#### Emergency Overview

Harmful if swallowed. Causes serious eye damage. May cause respiratory irritation.

#### Classification of the substance or mixture

|                                                    |            |
|----------------------------------------------------|------------|
| Acute Oral Toxicity                                | Category 4 |
| Serious Eye Damage/Eye Irritation                  | Category 1 |
| Specific target organ toxicity - (single exposure) | Category 3 |

#### Label Elements



**Signal Word**

**Danger**

## 2-Phenoxyethanol

**Hazard Statements**

H302 - Harmful if swallowed  
H318 - Causes serious eye damage  
H335 - May cause respiratory irritation

**Precautionary Statements****Prevention**

P261 - Avoid breathing dust/fume/gas/mist/vapors/spray  
P264 - Wash face, hands and any exposed skin thoroughly after handling  
P270 - Do not eat, drink or smoke when using this product  
P271 - Use only outdoors or in a well-ventilated area  
P280 - Wear protective gloves/protective clothing/eye protection/face protection

**Response**

P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing  
P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
P310 - Immediately call a POISON CENTER or doctor  
P330 - Rinse mouth

**Storage**

P403 + P233 - Store in a well-ventilated place. Keep container tightly closed

**Disposal**

P501 - Dispose of contents/ container to an approved waste disposal plant

**Physical and Chemical Hazards**

None identified.

**Health Hazards**

Harmful if swallowed. May cause respiratory irritation.

**Environmental hazards**

Contains no substances known to be hazardous to the environment or not degradable in waste water treatment plants. . Will likely be mobile in the environment due to its water solubility. The product is water soluble, and may spread in water systems.

**Other Hazards**

Toxic to terrestrial vertebrates. This product does not contain any known or suspected endocrine disruptors.

**SECTION 3. COMPOSITION/INFORMATION ON INGREDIENTS**

| Component                        | CAS No   | Weight % |
|----------------------------------|----------|----------|
| Ethylene glycol monophenyl ether | 122-99-6 | <= 100   |

**SECTION 4. FIRST AID MEASURES****General Advice**

If symptoms persist, call a physician.

**Eye Contact**

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.

**Skin Contact**

Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician.

**Inhalation**

Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur.

**Ingestion**

Clean mouth with water and drink afterwards plenty of water.

**Most important symptoms and effects**

Causes severe eye damage.

**Self-Protection of the First Aider**

Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.

**Notes to Physician**

Treat symptomatically.

**SECTION 5. FIRE-FIGHTING MEASURES****Suitable Extinguishing Media**

Water spray. Carbon dioxide (CO<sub>2</sub>). Dry chemical. Chemical foam.

**Extinguishing media which must not be used for safety reasons**

No information available.

**Specific Hazards Arising from the Chemical**

Thermal decomposition can lead to release of irritating gases and vapors.

**Protective Equipment and Precautions for Firefighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

**SECTION 6. ACCIDENTAL RELEASE MEASURES****Personal Precautions**

Ensure adequate ventilation. Use personal protective equipment as required.

**Environmental Precautions**

Should not be released into the environment.

**Methods for Containment and Clean Up**

Keep in suitable, closed containers for disposal. Soak up with inert absorbent material.

Refer to protective measures listed in Sections 8 and 13.

**SECTION 7. HANDLING AND STORAGE****Handling**

Wear personal protective equipment/face protection. Ensure adequate ventilation. Do not get in eyes, on skin, or on clothing. Avoid ingestion and inhalation.

**Storage**

Keep in a dry, cool and well-ventilated place. Keep container tightly closed.

**Specific Use(s)**

Use in laboratories

**SECTION 8. EXPOSURE CONTROLS/PERSONAL PROTECTION****Control Parameters****Exposure Controls**

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**Engineering Measures**

Ensure adequate ventilation, especially in confined areas. Ensure that eyewash stations and safety showers are close to the workstation location. Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source.

**Personal protective equipment**

**Eye Protection** Goggles (European standard - EN 166)

**Hand Protection** Protective gloves

| Glove material | Breakthrough time | Glove thickness | EU standard | Glove comments        |
|----------------|-------------------|-----------------|-------------|-----------------------|
| Nitrile rubber | See manufacturers | -               | EN 374      | (minimum requirement) |
| Neoprene       | recommendations   |                 |             |                       |
| Natural rubber |                   |                 |             |                       |
| PVC            |                   |                 |             |                       |

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatibility, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

**Skin and body protection** Wear appropriate protective gloves and clothing to prevent skin exposure

**Respiratory Protection** No protective equipment is needed under normal use conditions.

**Large scale/emergency use** Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced

**Small scale/Laboratory use** Maintain adequate ventilation

**Hygiene Measures** Handle in accordance with good industrial hygiene and safety practice.

**Environmental exposure controls** No information available.

**SECTION 9. PHYSICAL AND CHEMICAL PROPERTIES**

|                                     |                             |                                          |
|-------------------------------------|-----------------------------|------------------------------------------|
| <b>Appearance</b>                   | Colorless                   |                                          |
| <b>Physical State</b>               | Liquid                      |                                          |
| <b>Odor</b>                         | aromatic                    |                                          |
| <b>Odor Threshold</b>               | No data available           |                                          |
| <b>pH</b>                           | 8-9                         | 10g/100ml water/MeOH                     |
| <b>Melting Point/Range</b>          | 11 - 13 °C / 51.8 - 55.4 °F |                                          |
| <b>Softening Point</b>              | No data available           |                                          |
| <b>Boiling Point/Range</b>          | 237 °C / 458.6 °F           | @ 760 mmHg                               |
| <b>Flash Point</b>                  | 130 °C / 266 °F             | <b>Method -</b> No information available |
| <b>Evaporation Rate</b>             | No data available           |                                          |
| <b>Flammability (solid,gas)</b>     | Not applicable              | Liquid                                   |
| <b>Explosion Limits</b>             | No data available           |                                          |
| <b>Vapor Pressure</b>               | 0.04 mbar @ 20 °C           |                                          |
| <b>Vapor Density</b>                | 4.8 (Air = 1.0)             | (Air = 1.0)                              |
| <b>Specific Gravity / Density</b>   | 1.100                       |                                          |
| <b>Bulk Density</b>                 | Not applicable              | Liquid                                   |
| <b>Water Solubility</b>             | 30 g/l (20°C)               |                                          |
| <b>Solubility in other solvents</b> | No information available    |                                          |

**Partition Coefficient (n-octanol/water)**

|                                  |                          |
|----------------------------------|--------------------------|
| <b>Component</b>                 | <b>log Pow</b>           |
| Ethylene glycol monophenyl ether | 1.2                      |
| <b>Autoignition Temperature</b>  | 430 °C / 806 °F          |
| <b>Decomposition Temperature</b> | No data available        |
| <b>Viscosity</b>                 | 29 mPa s at 20 °C        |
| <b>Explosive Properties</b>      | No information available |
| <b>Oxidizing Properties</b>      | No information available |

|                          |           |
|--------------------------|-----------|
| <b>Molecular Formula</b> | C8 H10 O2 |
| <b>Molecular Weight</b>  | 138.17    |

**SECTION 10. STABILITY AND REACTIVITY**

|                                 |                                                           |
|---------------------------------|-----------------------------------------------------------|
| <b>Stability</b>                | Stable under recommended storage conditions.              |
| <b>Hazardous Reactions</b>      | None under normal processing.                             |
| <b>Hazardous Polymerization</b> | Hazardous polymerization does not occur.                  |
| <b>Conditions to Avoid</b>      | Incompatible products.                                    |
| <b>Materials to avoid</b>       | Strong oxidizing agents. Acid anhydrides. Acid chlorides. |

**Hazardous Decomposition Products** Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>).

**SECTION 11. TOXICOLOGICAL INFORMATION****Product Information****(a) acute toxicity;**

| <b>Component</b>                 | <b>LD50 Oral</b>          | <b>LD50 Dermal</b>        | <b>LC50 Inhalation</b>        |
|----------------------------------|---------------------------|---------------------------|-------------------------------|
| Ethylene glycol monophenyl ether | LD50 = 1850 mg/kg ( Rat ) | LD50 = 5 mL/kg ( Rabbit ) | LC50 > 0.057 mg/L ( Rat ) 8 h |

**(b) skin corrosion/irritation;** Based on available data, the classification criteria are not met

**(c) serious eye damage/irritation;** Category 1

**(d) respiratory or skin sensitization;**

|                    |                                                                  |
|--------------------|------------------------------------------------------------------|
| <b>Respiratory</b> | Based on available data, the classification criteria are not met |
| <b>Skin</b>        | Based on available data, the classification criteria are not met |

**(e) germ cell mutagenicity;** Based on available data, the classification criteria are not met

**(f) carcinogenicity;** Based on available data, the classification criteria are not met  
There are no known carcinogenic chemicals in this product

**(g) reproductive toxicity;** Based on available data, the classification criteria are not met

**(h) STOT-single exposure;** Category 3

|                                |                    |
|--------------------------------|--------------------|
| <b>Results / Target organs</b> | Respiratory system |
|--------------------------------|--------------------|

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(i) STOT-repeated exposure; Based on available data, the classification criteria are not met

## Target Organs

None known.

(j) aspiration hazard; Based on available data, the classification criteria are not met

## Other Adverse Effects

Symptoms / effects, both acute and delayed No information available

## SECTION 12. ECOLOGICAL INFORMATION

Ecotoxicity effects Do not empty into drains. .

| Component                        | Freshwater Fish                                                                                                    | Water Flea                            | Freshwater Algae                                | Microtox                                       |
|----------------------------------|--------------------------------------------------------------------------------------------------------------------|---------------------------------------|-------------------------------------------------|------------------------------------------------|
| Ethylene glycol monophenyl ether | LC50: = 366 mg/L, 96h static (Pimephales promelas)<br>LC50: 337 - 352 mg/L, 96h flow-through (Pimephales promelas) | EC50: > 500 mg/L, 48h (Daphnia magna) | EC50: > 500 mg/L, 72h (Desmodesmus subspicatus) | EC50 = 32.4 mg/L 5 min<br>EC50 = 880 mg/L 17 h |

## Persistence and Degradability

## Persistence

Persistence is unlikely.

## Bioaccumulative Potential

Bioaccumulation is unlikely

| Component                        | log Pow | Bioconcentration factor (BCF) |
|----------------------------------|---------|-------------------------------|
| Ethylene glycol monophenyl ether | 1.2     | No data available             |

## Mobility in soil

The product is water soluble, and may spread in water systems Will likely be mobile in the environment due to its water solubility Highly mobile in soils

## Endocrine Disruptor Information

This product does not contain any known or suspected endocrine disruptors

## Persistent Organic Pollutant

This product does not contain any known or suspected substance

## Ozone Depletion Potential

This product does not contain any known or suspected substance

## SECTION 13. DISPOSAL CONSIDERATIONS

## Waste from Residues/Unused Products

Waste is classified as hazardous. Dispose of in accordance with the European Directives on waste and hazardous waste. Dispose of in accordance with local regulations.

## Contaminated Packaging

Dispose of this container to hazardous or special waste collection point.

## Other Information

Waste codes should be assigned by the user based on the application for which the product was used. Do not empty into drains. Do not flush to sewer.

## SECTION 14. TRANSPORT INFORMATION

## Road and Rail Transport

Not Regulated

## IMDG/IMO

Not regulated

## 2-Phenoxyethanol

**IATA** Not regulated**Special Precautions for User** No special precautions required**SECTION 15. REGULATORY INFORMATION****International Inventories**

X = listed, China (IECSC), Europe (EINECS/ELINCS/NLP), U.S.A. (TSCA), Canada (DSL/NDSL), Philippines (PICCS), Japan (ENCS), Japan (ISHL), Australia (AICS), Korea (KECL).

| Component                        | The Inventory of Hazardous Chemicals (2015 Edition) | List of dangerous goods GB 12268 - 2012 | TCSI | IECSC | EINECS    | TSCA | DSL | PICCS | ENCS | ISHL | AICS | KECL     |
|----------------------------------|-----------------------------------------------------|-----------------------------------------|------|-------|-----------|------|-----|-------|------|------|------|----------|
| Ethylene glycol monophenyl ether | -                                                   | -                                       | X    | X     | 204-589-7 | X    | X   | X     | X    | X    | X    | KE-28257 |

**National Regulations****SECTION 16. OTHER INFORMATION**

**Creation Date** 16-Nov-2010  
**Revision Date** 05-Apr-2024  
**Revision Summary** SDS sections updated.

**Training Advice**

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

First aid for chemical exposure, including the use of eye wash and safety showers.

**Legend****CAS** - Chemical Abstracts Service**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances**PICCS** - Philippines Inventory of Chemicals and Chemical Substances**IECSC** - Chinese Inventory of Existing Chemical Substances**KECL** - Korean Existing and Evaluated Chemical Substances**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List**ENCS** - Japanese Existing and New Chemical Substances**AICS** - Australian Inventory of Chemical Substances**NZIoC** - New Zealand Inventory of Chemicals**WEL** - Workplace Exposure Limit**ACGIH** - American Conference of Governmental Industrial Hygienists**DNEL** - Derived No Effect Level**RPE** - Respiratory Protective Equipment**LC50** - Lethal Concentration 50%**NOEC** - No Observed Effect Concentration**PBT** - Persistent, Bioaccumulative, Toxic**TWA** - Time Weighted Average**IARC** - International Agency for Research on Cancer**PNEC** - Predicted No Effect Concentration**LD50** - Lethal Dose 50%**EC50** - Effective Concentration 50%**POW** - Partition coefficient Octanol:Water**vPvB** - very Persistent, very Bioaccumulative**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code**MARPOL** - International Convention for the Prevention of Pollution from Ships

**2-Phenoxyethanol**

**OECD** - Organisation for Economic Co-operation and Development  
**BCF** - Bioconcentration factor

**ATE** - Acute Toxicity Estimate  
**VOC** - (Volatile Organic Compound)

**Key literature references and sources for data**

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

**Disclaimer**

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**End of Safety Data Sheet**